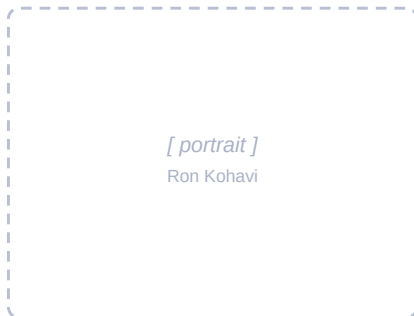


Ron Kohavi

Trustworthy Online Controlled Experiments (A/B Testing)

Computer scientist & data-science leader · ex-Microsoft, Amazon, Airbnb



"Getting numbers is easy; getting numbers you can trust is hard."

"Data trumps intuition — but only if the experiment is trustworthy."

1 WHY THIS THINKER MATTERS

When you run an A/B test in Unit 9, you are standing on Kohavi's work. He wrote the definitive guide to doing controlled experiments online without fooling yourself — the statistical conscience of modern product testing.

2 INTELLECTUAL FOUNDATION

Israeli-American computer scientist who led experimentation platforms at Microsoft, Amazon and Airbnb. With Tang and Xu he authored the standard reference on trustworthy online controlled experiments, cataloguing both the method and its many pitfalls.

3 FRAMEWORKS · TOOLS · MODELS

NAME	WHAT IT IS
Online Controlled Experiments	Randomised A/B tests on live products
Trustworthiness checks	Guarding against invalid or misleading results
OEC (Overall Evaluation Criterion)	One metric that captures long-term value
Twyman's law / surprising results	Extraordinary results usually mean errors
Experimentation platform	Infrastructure for testing at scale

4 HOW IT'S OPERATIONALIZED

- Define a single Overall Evaluation Criterion up front
- Randomise properly and check for sample-ratio mismatch
- Run tests to statistical significance, not to a hunch
- Treat surprising wins as likely bugs until verified
- Institutionalise experimentation with shared platforms

5 LIMITATIONS & CRITIQUES

- Requires large user volumes for power
- Short-run metrics may miss long-term effects
- Many subtle pitfalls invalidate naïve tests
- Local optimisation can miss step-change innovation
- Ethical and privacy constraints on live testing

6 ENDURING IMPACT

- Authored the standard text on online experiments
- Built experimentation platforms at scale at top tech firms
- Catalogued the pitfalls that make A/B tests trustworthy
- Raised the statistical bar for product decisions
- Direct anchor for BCOBM126 Unit 9 A/B testing

SELECTED WORKS

TRUSTWORTHY ONLINE
CONTROLLED
EXPERIMENTS

KOHAVI, TANG & XU · 2020

CONTROLLED
EXPERIMENTS ON THE
WEB

KOHAVI · 2009

ONLINE EXPERIMENTATION
AT MICROSOFT

case study

KOHAVI · 2009

PITFALLS OF ONLINE
EXPERIMENTS

KDD papers

KOHAVI · 2010s

REFERENCES (HARVARD · CITE THEM RIGHT)

- Kohavi, R., Tang, D. and Xu, Y. (2020) *Trustworthy Online Controlled Experiments*. Cambridge: Cambridge University Press.
- Kohavi, R. et al. (2009) 'Controlled experiments on the web', *Data Mining and Knowledge Discovery*, 18(1), pp. 140–181.